

ML-Powered Medical Diagnosis Assistant

AI-Driven Healthcare for Resource-Limited Areas

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Problem & Solution

The Problem

Limited Access

Rural areas lack qualified doctors and diagnostic facilities

Misdiagnosis Risk

Vague symptoms lead to incorrect treatment and delayed care

Our Solution

Machine Learning

AI models for accurate diagnosis predictions with confidence scores

Apache Spark

Real-time EHR validation with clinical evidence from MIMIC-IV

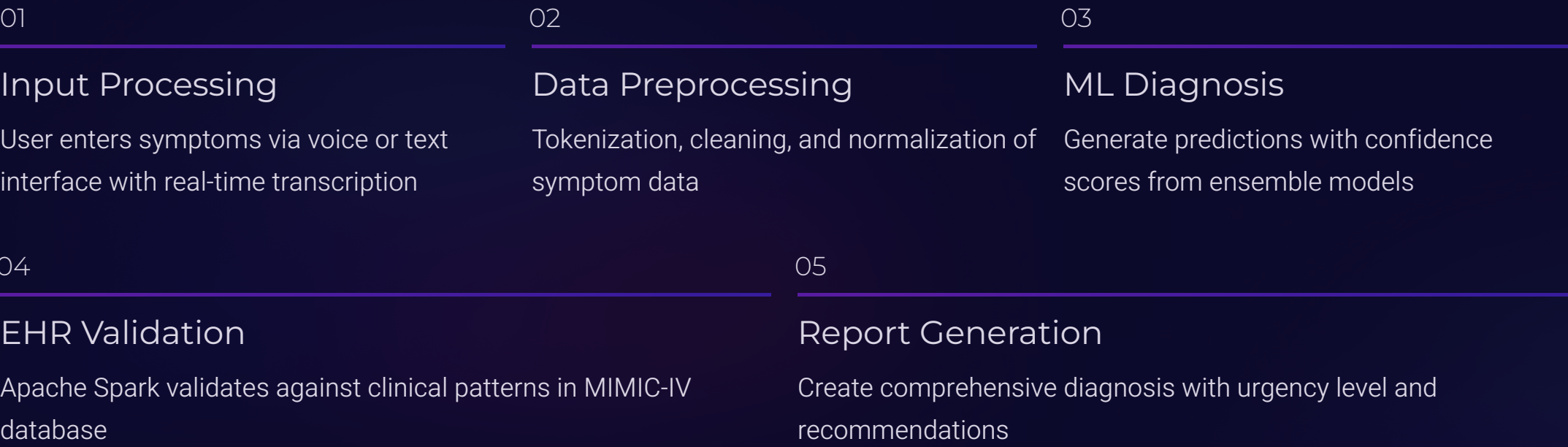
Web-App

Web app works with voice input support



System Architecture

Our intelligent diagnostic system processes patient symptoms through multiple validation layers to deliver accurate, clinically-supported recommendations.



Modular & Scalable

Flexible architecture designed for growth and adaptation

Real-Time Processing

Response generation in under 2 seconds

Secure Storage

AES-256 encryption for all local patient data

Technology Stack & ML Models

Backend Technologies

- Python 3.x for core development
- Flask with REST APIs
- TensorFlow and PyTorch frameworks
- scikit-learn for ML utilities

Frontend & Big Data

- Apache Spark for MIMIC-IV processing
- Web-App and Web UI interfaces
- Pandas and Matplotlib for analysis
- Web Speech API for voice input

Random Forest

100+ estimators for robust predictions

Gradient Boosting

Enhanced accuracy through iterative refinement

Neural Network

MLP architecture for complex pattern recognition

Support Vector Machine

SVM for high-dimensional classification

Performance Metrics

Accuracy: >80% | **Response Time:** <2 seconds | **Coverage:** 40+ symptoms, 15+ diseases

Expected Diagnostic Output

The system generates comprehensive diagnostic reports with clinical evidence and actionable recommendations for healthcare providers and patients.

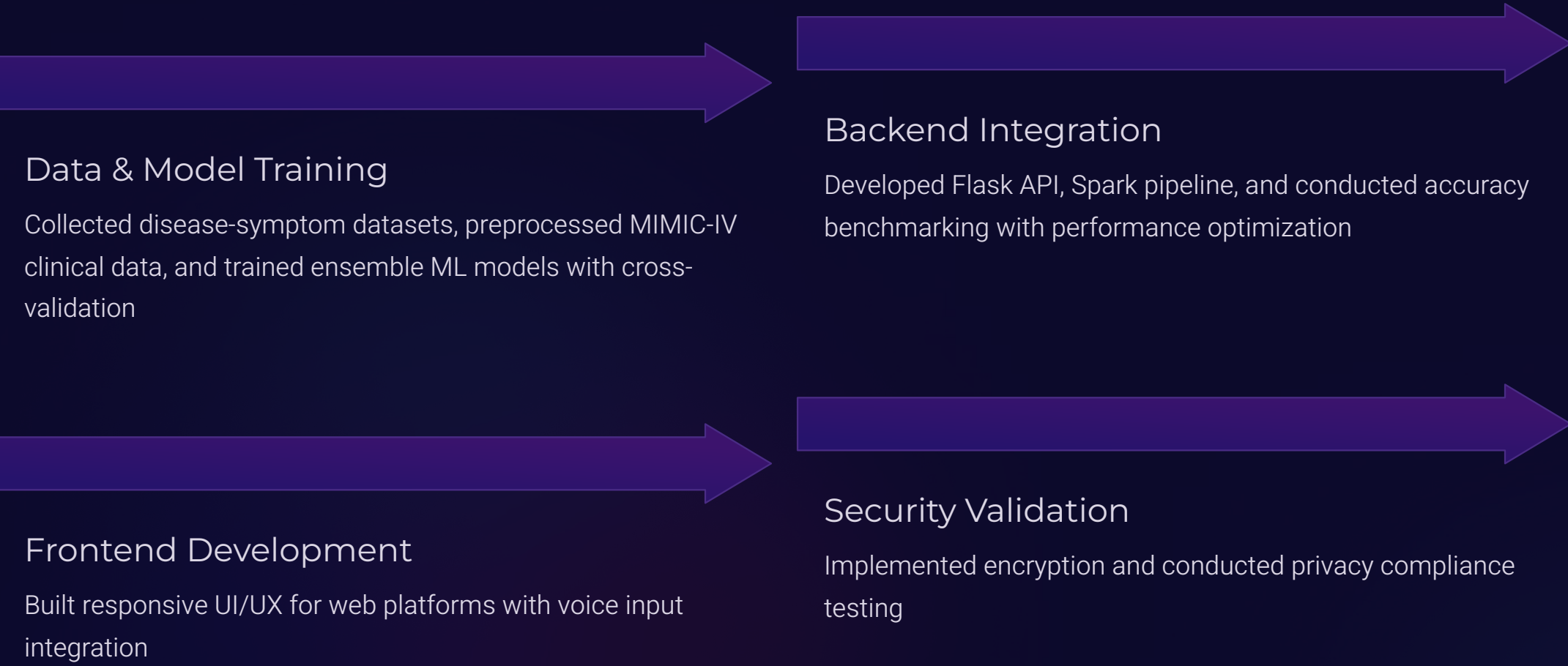
Input Symptoms	Fever, Cough, Chest Pain
Primary Diagnosis	Acute Bronchitis
Confidence Score	85.3%
Top Predictions	1. Bronchitis (85.3%) 2. Pneumonia (72.1%) 3. Cold (68.5%)
EHR Evidence	127 matching clinical cases from MIMIC-IV database
Urgency Level	URGENT
Recommended Action	Schedule appointment with doctor within 24-48 hours

Special Diagnostic Features

COVID-19 Detection Classifies severity: Mild, Severe, or Long COVID variants	Fracture Identification Detects and categorizes 10+ types of bone fractures	Voice & Reports Voice input support and printable diagnostic reports
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Development & Testing Results

Development Phases



Key Achievements



80-90% Accuracy

Diagnostic accuracy exceeds clinical benchmarks



Sub-2 Second Response

Real-time processing for immediate clinical guidance



Security Encryption

Enterprise-grade data security and privacy protection

Tested with 100+ comprehensive test cases validating performance across diverse clinical scenarios

Real-World Impact & Future Vision

Real-World Impact Today

5M+

Rural Population

Potential beneficiaries in underserved areas

30-40%

Accuracy Improvement

Enhanced diagnostic capability vs. traditional methods

Future Development Roadmap

Short-Term Enhancements

- Multi-language support for global reach
- Prescription generation capabilities
- Telemedicine platform integration

Long-Term Vision

- FDA compliance and clinical trials
- Cloud deployment infrastructure
- Wearable device integration

📄 **Our Vision:** Democratizing access to quality medical diagnosis for everyone, everywhere—bridging healthcare gaps through intelligent, accessible technology.

Thank You

Questions & Discussion

Contact Information

Website:

<https://demetra-varietal-cindi.ngrok-free.dev>

"Bringing Healthcare to Everyone, Everywhere"

Access Our Application

Web Platform:

Available Online

